

Aishwarya Govindappa Mahadev

🔗 GitHub: github.com/Aishwarya-G-M 🌟 Featured Project: Secure LLM Gateway (LLM Guard, LLM models, LLM Proxy)

✉ aishwarya.govindappamahadev@gmail.com ☎ +918807836124 🌐 linkedin.com/in/aishwarya-govindappa-mahadev

EXPERIENCE

Senior Software Engineer (Product & Platform Engineering),

09/2023 – Present | Bangalore, India

Altimetrik (Client: Airbnb)

- As a part of Search Infra team's initiatives for Airbnb's **LLM-powered AI Search platform**, I was involved in designing and **driving adoption** of a shared caching framework that standardised **LLM integration patterns** across multiple search services. Provisioned and managed **Redis infrastructure** using **Terraform**, introduced a unified Redis client abstraction replacing legacy Redisson/Jedis implementations, and enabled **scalable, maintainable** cache integrations through **reusable platform components**.
- Performed **platform integration** activities for AI search services including **service mesh** configuration, **service discovery** onboarding, cross-service whitelisting, and **automated integration testing**, ensuring **secure** and **reliable communication**.
- Supported deployment and operation of **containerised services** running on **Kubernetes**, collaborating on **infrastructure provisioning**, service onboarding, **configuration management**, and **production support** activities across AI Search platforms.
- Implemented **multi-layer caching strategies** across endpoint, guardrail, and LLM invocation layers for **high-traffic services**, improving **P99 latency**, reducing Redis access overhead, and enabling **safe production rollouts** through **feature-flag-driven controls**, cache policies, and TTL management.
- Enhanced **AI platform security** and **governance controls** by implementing **jailbreak-detection**, **guardrail enforcement**, and **policy enforcement mechanisms** that prevented discriminatory and policy-violating prompts from bypassing safety controls.
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- Built the first end-to-end **integration testing framework** for **Knowledge Graph pipelines** spanning Flink, Gremlin, JanusGraph, and Mussel services, enabling **automated validation** of complex **distributed data flows** prior to **production deployment**.
- Modernised and optimised **Spark-based data pipelines** and **Airflow workflows**, improving **reliability**, **runtime efficiency**, and **upgrade readiness** through enhanced **test automation** and platform modernisation efforts.
- Improved **build performance**, **maintainability**, and **engineering standards** of large-scale Java/Scala monorepo services through dependency cleanup, legacy code refactoring, and **build-system optimization** initiatives.

Advanced Software Engineer, Zuhlke (Client: Credit Suisse)

08/2021 – 08/2022 | Singapore, Singapore

- Co-developed and owned multiple production-grade RESTful microservices within Credit-Suisse's **Wealth Management platform**, powering critical portfolio, trading, and advisory workflows for financial assets. Designed **scalable APIs**, implemented complex business logic, and ensured **secure, low-latency integration** with upstream and downstream **enterprise systems**.
- Improved **service resilience** and **operational continuity** of a critical trading workflow by designing and implementing a **fault-tolerant fallback mechanism** that eliminated **single points of failure**, reduced dependency on third-party services, and maintained business-critical functionality during external outages.
- Provided **production** support through incident investigation, root-cause analysis, debugging, and operational readiness checks, enabling stable releases.
- Contributed to engineering best practices through API design reviews, code reviews, technical documentation, and knowledge sharing activities, helping improve software quality, maintainability, and team productivity across the platform.

Software Engineer, Garmin

10/2019 – 12/2020 | Olathe, USA

- Implemented **RESTful services**, strictly enforcing secure architecture design patterns, data encryption, API versioning, and rigorous access controls.
- Safeguarded production-critical infrastructure through end-to-end lifecycle ownership, ensuring security posture resilience during major system migrations and feature rollouts.
- Mitigated operational risk** by leading real-time troubleshooting for manufacturing discrepancies and conducting rigorous code reviews to eliminate security flaws before deployment.
- Optimized threat visibility and system health by setting up **Grafana dashboards** to track anomalies, and defining critical service-level security metrics.
- Established secure engineering baselines** across the team by authoring technical documentation, facilitating Scrum ceremonies, and mentoring new engineers.

Associate Consultant, Oracle Financial Software Services (Client: State Street)

10/2013 – 07/2015 | India

- Supported the large-scale migration of Oracle 9 DB to Oracle Exadata; Developed scripts to automate recurring operational tasks and significantly improving efficiency and reducing manual effort during the transition.
- Provided critical production support during releases, performing troubleshooting, debugging, and live validation to minimize downtime and ensure smooth deployment cycles for business-critical systems.
- Executed comprehensive sanity checks in SIT and UAT environments, validating data consistency and service stability.

EDUCATION

MSc in Computer Science, *Texas A&M University*

2018 – 2019 | Commerce, USA

SKILLS

Cloud & Infrastructure — AWS | Kubernetes | Docker | Terraform | OpenShift | Vault | Service Mesh | Infrastructure as Code (IaC)

Platform Engineering — Distributed Systems | Service Discovery | Observability | Production Operations | CI/CD | Platform Architecture | Reliability Engineering

Languages — Java | Scala | Python

Backend & Frameworks — Spring Boot | Fast API | Microservices | Monoliths and Monorepos | API Design (RESTful, GraphQL, MVC, SOA)

Data Platforms & Streaming: — Spark | Flink | Airflow | Kafka | Superset

Security & Governance — Application Security | LLM Guardrails | Secure Service Communication

Certificates

Generative AI for Everyone - DeepLearning.AI  — (Sept 2025)

Prompt Engineering (Airbnb, DeepLearning.AI, Altimetrik)  — (Sept 2025)

Introduction to Responsible AI - Google Cloud  — (Sept 2025)

Trustworthy Generative AI - Vanderbilt University  — (Sept 2025)